A Curved-Spacetime Vacuum Kernel and the Gravitational-Sector Closure of Event-Density Theory

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Abstract

We extend the Event-Density (ED) program from its Phase-2 + Arc N quantum-sector closure to a gravitational-sector structural foundation. Phase-3 addresses the question: *does ED's primitive-level structure force, admit, or forbid gravitational coupling, dynamical curvature, and cosmological structure?* Through a six-stage analysis (GR.0 scoping, GR.1 catalogue of admissible curvature-coupling structures, GR.2 FORCED evaluation, GR.3 REFUTED evaluation, GR.4 cosmological implications, GR.5 synthesis), we establish **Theorem GR1 (V1 with Synge World Function forced)**: the vacuum-response kernel in curved spacetime is structurally forced at primitive level to take the form $K_{\text{vac}}^{\text{curved}}(x, x') = K_{\text{vac}}^{\text{prim}}(\sigma(x, x')/\ell_{\text{ED}}^2)$ with $\sigma(x, x')$ Synge's world function, via a Hadamard-parametrix-style extension of Arc N's Theorem N1 to curved spacetime. Three FORCED-conditional cascading items emerge (free-chain geodesic worldlines via eikonal limit, curvature-dependent cross-chain correlations, Λ-integral existence). Seven REFUTED determinations under the C1/C2/C3/locality + Phase-2-theorem framework bound the admissibility space — including the unconditional refutation of curvature-dependent Case-P/Case-R distinctions by spin-statistics applying pointwise at every event regardless of curvature. GR-4A (Einstein-equation emergence from primitives) remains NOT REFUTED BUT SPECULATIVE: ED admits Einstein equations as effective-theory addition without deriving them from primitives. ED-Phys-10's acoustic-metric-only baseline is preserved throughout. Stage GR.4 establishes a coherent cosmological framework with four substantive structural channels — Λ as V1-kernel integral with finite primitive-level magnitude (structurally dissolving the cosmological-constant divergence-form puzzle while Λ's magnitude remains INHERITED), V1-induced dispersion-relation modifications, V5-mediated cosmological correlations, and curvature-vacuum coupling — and four empirical-signature routes (UHECR timing, GRB photon timing, gravitational-wave dispersion, cosmological-correlation persistence). Phase-3 lands in the partial-GR-induce quadrant, paralleling Arc M's H1-dominant pattern. Theorem GR1 takes its place as ED's first gravitational-sector structural theorem and the ninth FORCED theorem in the ED structural inventory, completing the six-layer structural foundation while preserving the *form-FORCED, value-INHERITED* methodology.

1 Introduction

The Event-Density (ED) program builds quantum and gravitational physics as forced structural consequences of a small primitive stack on a 3+1-dimensional event manifold. Phase-1 [1] derived non-relativistic single-particle quantum mechanics. Phase-2 extended this through three connected arcs: Arc R [2] (Klein–Gordon, spin–statistics, Cl(3,1) frame, Dirac equation); Arc M [3] (mass structure: form FORCED, values INHERITED); Arc Q [4] (QFT extension: GRH unconditional FORCED, canonical (anti-)commutation FORCED, UV-FIN FORCED). Arc N [5] closed the kernel-level memory-structure layer with Theorem N1 (V1 finite-width vacuum memory kernel FORCED).

Phase-3 opens the gravitational sector. The driving question:

(Q-Phase-3) Does ED's primitive-level structure force, admit, or forbid gravitational coupling, dynamical curvature, and cosmological structure?

Phase-3 closes with the answer: **(GR-induce) partial**. ED admits structural curvature-coupling channels — most cleanly via Theorem GR1's curved-spacetime extension of V1 — but does not derive full Einstein dynamics from primitives.

This paper presents the six-stage closure: GR.0 scoping [6], GR.1 catalogue [7], GR.2 FORCED evaluation [8], GR.3 REFUTED evaluation [9], GR.4 cosmological implications [10], GR.5 synthesis [11].

The headline verdict: **ED forces a curved-spacetime vacuum kernel, forces geodesic free-chain motion conditionally, forces Λ to be finite and kernel-integral-defined, admits but does not force curvature-dynamics equations, and provides a coherent cosmological framework with testable signatures.** All Phase-2 + Arc N theorems are preserved; ED-Phys-10's prior closure [12] is preserved as the structural ceiling.

The paper is organised as follows. Section 2 reviews the ED structural inventory and Phase-3's position. Section 3 presents the roadmap. Section 4 states and proves Theorem GR1. Section 5 presents REFUTED and ADMISSIBLE structures. Section 6 evaluates cosmological implications. Section 7 states the final Phase-3 verdict. Section 8 outlines the Phase-4 outlook.

2 Background and Position in the ED Program

2.1 ED structural inventory (post-Arc N)

Phase-3 inherits eight FORCED structural theorems:

#	Theorem	Source / Type
1	Spin–statistics $\eta = (-1)^{2s}$	R.2.5 / Theorem-level
2	Cl(3,1) frame uniqueness	R.2.4 / Algebra-level
3	Anyon prohibition in 3+1D	R.2.3 / Topology-level
4	Dirac equation emergence	R.3 / Dynamical-equation-level
5	GRH unconditional FORCED	Q.1 + Q.2/3/7/8 / Existence-level
6	Canonical (anti-)commutation FORCED	Q.7 / Operator-level
7	UV-FIN FORCED	Q.8 / Bound-level
8	V1 finite-width vacuum kernel FORCED	N.2 + N.5 / Memory-kernel-level

Theorem GR1 becomes the ninth FORCED theorem and the first at the curved-spacetime-kernel-level.

2.2 ED-Phys-10 prior closure as structural ceiling

The ED-Phys-10 kinematic-curvature arc [12] established the acoustic effective metric for bandwidth-mode perturbations, causal-cone structure, and analogue-horizon structures. It explicitly closed *without*: Einstein equations, Schwarzschild solution, Newtonian-gravity emergence at long-wavelength, or fine-structure constant α from primitives. Phase-3 must produce results consistent with these guardrails or explicitly relax them.

2.3 What Arc N changes

Arc N's Theorem N1 supplies a primitive-level structural mechanism for non-Markovian vacuum response. In flat space, V1's kernel depends on the Lorentz invariant $(x-x')^2$. In curved spacetime, the natural generalisation replaces this with Synge's world function $\sigma(x, x')$ — substantively new structural input enabling Theorem GR1.

2.4 Why gravitational coupling is a Phase, not an Arc

Arcs R, M, Q, N each addressed a single coherent structural question within the quantum sector. Phase-3 spans a different conceptual layer entirely: gravitational coupling and cosmological structure. The shift is from quantum-sector to gravitational-sector content, requiring extension to curved spacetime, integration with the kinematic acoustic-metric framework, and consideration of cosmological-scale phenomenology.

2.5 Six-layer ED structural foundation

Post-Phase-3, the ED structural foundation spans six layers:

Layer	Content
Phase-1	Non-relativistic QM
Phase-2 Arc R	Relativistic kinematics + dynamics
Phase-2 Arc M	Mass structure
Phase-2 Arc Q	QFT extension
Phase-2 Arc N	Memory-kernel layer
Phase-3	**Gravitational sector**

3 Phase-3 Roadmap and Methodology

Phase-3 proceeds through six substages: GR.0 scoping [6]; GR.1 catalogue [7] (20 admissible curvature-coupling structures across GR-1 bandwidth-curvature, GR-2 vacuum-curvature, GR-3 worldline-curvature, GR-4 curvature-dynamics sectors); GR.2 FORCED evaluation [8] (Theorem GR1 unconditionally FORCED; three cascading FORCED-conditional; GR-4A SPECULATIVE); GR.3 REFUTED evaluation [9] (seven REFUTED determinations); GR.4 cosmological implications [10]; GR.5 synthesis [11].

Methodological discipline preserved throughout: clear separation of FORCED/CANDIDATE/REFUTED/ADMISSIB honest expected-verdict prior (45% partial GR-induce / 30% GR-add / 15% GR-emerge / 10% surprise); *form-FORCED, value-INHERITED* framing; ED-Phys-10 guardrails preserved.

4 Theorem GR1: Curved-Spacetime V1 Kernel forced

4.1 Statement

Theorem 4.1 (V1 with Synge World Function FORCED, GR1)**.** *In curved spacetime where unique geodesics connect any two events x, x', the vacuum-response kernel is structurally FORCED at primitive level to take the form*

$$K_{\text{vac}}^{\text{curved}}(x, x') = K_{\text{vac}}^{\text{prim}}(\sigma(x, x')/\ell_{\text{ED}}^2), \tag{1}$$

with $\sigma(x, x')$ Synge's world function (one-half squared geodesic distance), $K_{\text{vac}}^{\text{prim}}$ a kernel of admissibility class identical to the Theorem N1 flat-space class (finite, non-zero, decaying, Lorentz-scalar, sub-power-law-2), and ℓ_{ED} the primitive event-discreteness scale.

4.2 Three-leg proof sketch

The forcing argument extends Theorem N1's three-leg flat-space argument [5] to curved spacetime via Hadamard-parametrix construction.

Leg 1 — UV-FIN is metric-independent. Arc Q's UV-FIN argument (Theorem Q3 [4]) operates by primitive-level finiteness of multi-chain participation integrals via Primitive 01 + 13 + 04 jointly. Each primitive is **metric-independent**: they operate on the underlying event manifold regardless of curvature. UV-FIN extends to curved spacetime unchanged.

A δ-function vacuum response in curved spacetime, $K_{\text{vac}}^{\text{curved}} \propto \delta_g^{(4)}(x, x')$, has Fourier transform with respect to a local-inertial-frame at x that produces constant frequency-weighting — exactly as in flat space. UV-amplification pathology applies pointwise at every event. The δ-curved limit is REFUTED for the same reason V1-δ is REFUTED in flat space.

Leg 2 — General covariance + locality. A constant kernel $K_{\text{vac}}^{\text{curved}}(x, x') = c_\infty$ for all x, x' implies infinite correlation length on the curved manifold. Generally-covariant locality (extension of Primitive 11 to curved spacetime: commitment events remain point-events) forbids constant kernels. The infinite-width limit is REFUTED by C1 + locality.

Leg 3 — 3+1D power-counting applies pointwise. The 3+1D power-counting argument from Theorem N1 ($\alpha < 2$ for power-law kernels) operates locally — it depends on the dimensionality of spacetime at the event-resolution scale, not on global curvature. The argument applies pointwise to $\sigma(x, x')/\ell_{\text{ED}}^2$ in place of $(x - x')^2/\ell_{\text{ED}}^2$.

Synthesis. The three legs jointly force $K_{\text{vac}}^{\text{curved}}$ to satisfy finite-width, decaying-support, sub-power-law-2 conditions in $\sigma(x, x')$. The Synge-world-function form is the unique generally-covariant scalar two-point function reducing to flat-space $\frac{1}{2}(x - x')^2$ in the flat limit. The Hadamard parametrix construction [13, 14] supplies the standard mathematical infrastructure. □

4.3 Dependencies

Theorem GR1 depends on:

- Primitive 01 (event-discreteness, metric-independent): natural-cutoff scale ℓ_{ED}.
- Primitive 04 (bounded bandwidth, metric-independent): amplitude content.
- Primitive 06 (general covariance): C1 constraint.
- Primitive 11 (commitment-event locality): C1-locality refinement.
- Primitive 13 (finite proper-time intervals, generalised to affine-parameter intervals).
- Theorem N1: flat-space prerequisite.
- Theorem Q3 (UV-FIN): primitive-level UV finiteness.
- ED-Phys-10: supplies geodesic structure where no proposed dynamical metric exists.

4.4 Bounded admissibility class

Stage GR.3 limit refutations bound Theorem GR1's admissibility class:

- **GR-2A-δ (zero-width limit) refuted by C3** — bounds the FORCED class from below.
- **GR-2A-∞ (infinite-width limit) refuted by C1 + locality** — bounds the FORCED class from above.

The bounds are derived consequences of C1 + C3 in their curved-spacetime extensions — not additional postulates.

4.5 Cascading forced-conditional items

Theorem GR1 propagates FORCED-conditional status to three secondary catalogue items:

- **GR-3A (free-chain geodesic worldlines):** KG/Dirac eikonal limit yields geodesic motion. FORCED-conditional on acoustic-metric identification.
- **GR-2D (curvature-dependent cross-chain correlations):** V5 inherits Synge-world-function structure under Theorem GR1.
- **GR-4D (curvature-induced vacuum backreaction, existence sense):** $\Lambda_{\text{primitive}}(x) \sim \int K_{\text{vac}}^{\text{curved}}(x, x')\rho_{\text{vac}}(x')d^4x'$ FORCED-conditional.

4.6 Significance

Theorem GR1 is **ED's first gravitational-sector structural theorem** and the **ninth forced theorem** in the ED inventory.

5 refuted and admissible Structures

5.1 refuted determinations

ID / Sub-case	Constraint	Brief reason
GR-1E (Case-P/R curvature-dep.)	C2	Spin-statistics applies pointwise; metric-independent
GR-1D (TYPE-mixing Fierz)	C2	Fierz TYPE classification is rule-type data
GR-3B (Case-R-delay)	C2	Pauli exclusion violation in delay window
GR-2A-δ (zero-width)	C3	δ-distribution amplifies UV
GR-2A-∞ (infinite-width)	C1 + locality	Constant kernel breaks locality
GR-3A-non-geodesic	eikonal limit	KG/Dirac eikonal forces geodesics
GR-2D large-amp/long-range	C1, C3	Cascade refutation

The pattern: 2 C1, 3 C2, 2 C3 violations + 1 cascade + 1 Phase-2-theorem violation. Constraint framework exhaustive.

5.2 Theorem GR1 admissibility class bounded both ways

The two GR-2A limit refutations together establish that Theorem GR1's admissibility class is bounded: kernels must have finite, non-zero, decaying width. The pattern parallels Theorem N1's V1 admissibility-class bounding in flat space.

5.3 admissible-not-forced structures

13 catalogue items + 3 admissible sub-cases + 4 FORCED-related items pass Stage GR.3 unrefuted:

ID	Description	Influences
GR-1A	Curvature-dep. bandwidth evolution	Arc M coupling refinement
GR-1B	Curvature-modified σ_τ shift	Arc M, structure formation
GR-1C	Curvature-dep. bandweights	Arc M
GR-1D, same-TYPE	Same-TYPE Fierz corrections	Arc M within-class
GR-2B	Curvature-dep. kernel width	Λ, dispersion
GR-2C	Curvature-dep. amplitude	Λ scaling
GR-2D, bounded	Bounded cross-chain correlations	Cosmological correlations
GR-2E	Curvature-dep. vacuum polarisation	Arc Q.5 cosmological
GR-3A	Free-chain geodesics (cond.)	Test-particle motion
GR-3B, non-Case-R-delay	Quantitative threshold-shift	Arc Q.7 cosmological
GR-3C	Curvature-dep. adjacency graphs	Multi-chain dynamics
GR-3D	Curvature-dep. memory kernels	Arc N cosmological
GR-3E	Curvature-dep. null-cone deformation	ED-Phys-10 baseline
GR-4B	Non-local curvature equations	Alternative to GR-4A
GR-4C	Acoustic-metric-only baseline	ED-Phys-10 preserved
GR-4D, existence	Vacuum backreaction (FORCED-cond.)	Λ structure
GR-4E	Curvature-dep. UV-FIN	Phase-4 work

5.4 GR-4A unique status

GR-4A (Einstein-like equations) is **not refuted but speculative**: ADMISSIBLE as effective-theory addition; not derivable from primitives. ED-Phys-10's "no Einstein equations from primitives" preserved.

5.5 Bounded curvature-coupling space

Admissible curvature-coupling structure in ED is bounded, not arbitrary. C1 forbids frame-dependent and infinite-correlation-length structures; C2 forbids spin-statistics-violating sub-cases; C3 forbids insufficient short-time decay; eikonal-limit forbids non-geodesic free-chain motion.

6 Cosmological Implications

Stage GR.4 [10] establishes four substantive cosmological structural channels.

6.1 Λ as V1-kernel integral

$$\Lambda_{\text{primitive}}(x) \sim \int K_{\text{vac}}^{\text{curved}}(x, x') \cdot \rho_{\text{vac}}(x')\, d^4x'. \tag{2}$$

Form FORCED-conditional via GR-4D + Theorem GR1; magnitude INHERITED. ED structurally **dissolves** the cosmological-Λ divergence-form puzzle but does not predict $\Lambda \approx 10^{-122}$.

In FLRW backgrounds, $\Lambda(t)$ admissible structurally without forcing $w(z) \neq -1$. Scale-stratified Λ structure emerges between primitive-level UV-FIN protection and effective-theory long-wavelength behaviour.

6.2 V1-induced dispersion-relation modifications

$$\omega^2 = c^2|\mathbf{k}|^2 + \delta\omega^2(\mathbf{k}; K_{\text{vac}}). \tag{3}$$

Form FORCED; magnitudes INHERITED.

6.3 V5-mediated cosmological correlations

$$\langle \delta b^{\text{env}}_{K_1}(x_1) \cdot \delta b^{\text{env}}_{K_2}(x_2) \rangle = K_{\text{cross}}(\sigma(x_1, x_2)). \tag{4}$$

Existence FORCED-conditional via cascade; amplitudes/ranges INHERITED.

6.4 Curvature-vacuum coupling

Theorem GR1 itself is the curvature-vacuum coupling result. Refinements GR-2B/C/E ADMISSIBLE.

6.5 CP-phase enrichment carryover

C-N4.1 (vacuum-mediated additional CP phases beyond Jarlskog count for ≥ 3 generations) carries forward as cosmological CANDIDATE: modified baryogenesis channels, CP-violation correlations across cosmological scales, CP-violating vacuum-polarisation contributions. Remains a CANDIDATE.

6.6 Empirical-signature framework

Four observational channels:

- **UHECR timing** ($\sim 10^{19}$ eV scales) — V1-induced dispersion offsets.
- **GRB photon timing** (TeV scales; existing Fermi-LAT bounds).
- **Gravitational-wave dispersion** (LIGO/Virgo/KAGRA bounds).
- **Cosmological-correlation persistence** (LSS power-spectrum modifications).

Each route inherits V1/V5 kernel structure; specific testability INHERITED.

6.7 Distinguishing structural vs inherited

Structurally forced: existence of finite Λ; existence of cross-chain correlations; geodesic free-chain motion; bounded V1 admissibility class; V1-induced dispersion modification form.

Inherited: Λ magnitude; H_0, $w(z)$; G; specific kernel widths and forms; correlation amplitudes/ranges; power-spectrum corrections; inflation/dark-matter/baryogenesis specifics; generation count.

7 Final Phase-3 Verdict

1. **ED forces a curved-spacetime vacuum kernel** — Theorem GR1 FORCED.
2. **ED forces geodesic free-chain motion (conditional)** — GR-3A FORCED-conditional via eikonal limit.
3. **ED forces Λ to be finite and kernel-integral-defined** — GR-4D existence FORCED-conditional; magnitude INHERITED; **cosmological-Λ divergence-form structurally dissolved**.
4. **ED admits but does not force curvature-dynamics equations** — GR-4A NOT REFUTED BUT SPECULATIVE; ED-Phys-10 baseline preserved.
5. **ED preserves all Phase-2 + Arc N theorems** — no prior verdict overturned; three Phase-2 closures structurally enriched (Arc M σ_τ, Arc Q.5 vacuum polarisation, Arc Q.8 cosmological-Λ).

6. **ED provides a coherent cosmological framework with testable signatures** — four substantive structural channels + four empirical-signature routes.

7. **Phase-3 lands in the partial-GR-induce quadrant** — paralleling Arc M's H1-dominant pattern.

8 Outlook (Phase-4 Preview)

Phase-3's closure positions ED through the gravitational-sector-coupling layer. Phase-4 (or extended Phase-3 work) opens with several open items:

8.1 ED-GR action candidates

Three candidate forms: effective Einstein-Hilbert + V1-kernel-corrected source (SPECULATIVE); non-local effective action via Theorem GR1 cascades (ADMISSIBLE under GR-4B); acoustic-metric-only effective action (ED-Phys-10 baseline preserved). Promotion to FORCED requires either new structural input or a derivation strategy not yet identified.

8.2 Empirical tests

Four observational channels (UHECR timing, GRB photon timing, gravitational-wave dispersion, cosmological-correlation persistence) await sharpening of V1/V5 kernel-parameter inference.

8.3 Kernel parameter inference

Combining empirical bounds from the four observational channels could constrain V1 kernel parameters (width ℓ_{ED}, decay form, multi-scale weights) and V5 correlation parameters. Joint analysis could in principle infer kernel structure at the percent level.

8.4 GR-4A promotion question

The most consequential open structural question: can a primitive-level argument promote GR-4A from SPECULATIVE to FORCED? Three plausible routes (new primitive introduction, derivation via novel argument, effective-theory upgrade via UV-FIN) all require structural content not yet identified.

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